JAGHDHEEP M

Data Scientist | Machine Learning Engineer

Bengaluru, India • +91 93423 22599 • jaghdheep@gmail.com
LinkedIn: linkedin.com/in/jaghdheepm • **GitHub:** github.com/JAGHDHEEP • **Available: Immediately**

PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, VIT) and published IoT researcher who builds machine learning systems that reach production, not just notebooks. Built a LangChain **RAG assistant** that answers grounded questions over research papers; trained and shipped an **XGBoost** clinical risk model (0.87 ROC-AUC) as a Flask app with SHAP interpretability; engineered real-time **YOLOv8** computer vision for industrial safety; and automated **ETL pipelines in Python and SQL** as a data engineering intern. Strong across the full stack of a data problem — pipelines, statistical modeling, evaluation, and deployment — and mentors an AI/ML community of 100+ students.

PROFESSIONAL EXPERIENCE

Data Engineer Intern — Venolin Technology *May 2025 – July 2025*

- Replaced a fully manual HR and financial reconciliation process with automated **ETL pipelines in Python and SQL**, turning a recurring hand-run task into a repeatable job the team could trust every reporting cycle.
- Wrote reusable **Pandas** cleaning and transformation modules that normalised raw, multi-source records into validated SQL schemas, catching malformed data at ingest instead of in downstream reports.
- Worked directly with cross-functional stakeholders to convert business questions into pipeline requirements, so the data landed in the shape the reporting team actually needed.

DATA SCIENCE & MACHINE LEARNING PROJECTS

Research Buddy — RAG Question-Answering Assistant for Academic Papers | *Python, LangChain, LLM, Embeddings, Vector Database*

- Built a **retrieval-augmented generation (RAG)** assistant that answers natural-language questions directly from uploaded research papers, replacing manual skimming of long PDFs with targeted, source-backed answers.
- Engineered the full retrieval pipeline in **LangChain** — document parsing, overlapping chunking, embedding generation, and vector similarity search — so only the highest-relevance passages enter the LLM context window.
- Constrained the model to respond strictly from retrieved context, sharply reducing hallucinations and making every answer traceable back to the source passage that produced it.

Clinical Risk Prediction — End-to-End ML System with Interpretability | *Python, scikit-learn, XGBoost, SHAP, Flask, SQLite*

- Ran exploratory data analysis and statistical hypothesis testing to drive feature selection, then trained and benchmarked four classifiers (Logistic Regression, Random Forest, SVM, XGBoost); **XGBoost won at 84% accuracy and 0.87 ROC-AUC**.
- Optimised hyperparameters via GridSearchCV and confirmed the result held under k-fold cross-validation and Voting/Stacking ensembles; applied **SHAP** so every prediction ships with the clinical factors that drove it — not a black box.
- Shipped the model as a Flask application with user authentication, CSV upload for batch scoring, and SQLite-backed prediction history, owning the full lifecycle from raw CSV to working service.

Factory Hazard Mitigation System — Real-Time Computer Vision Safety | *YOLOv8, OpenCV, Ultrasonic Sensors, Microcontrollers*

- Built a real-time worker–machine proximity monitor that fuses **YOLOv8** object detection with ultrasonic distance readings, cross-checking two independent signals to cut false alarms on a noisy factory floor.
- Wired threshold breaches straight to microcontroller-driven alerts, giving workers hands-free hazard warnings the instant they entered an unsafe zone.

Samachar — Personalized News Platform | *Python, Tornado, Streamlit, Firebase, Google OAuth2*

- Built a full-stack news product with Google OAuth2 sign-in and Firebase-backed preferences, plus an automated pipeline aggregating articles from external APIs into scheduled per-user email digests.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning & Statistics: Logistic Regression, Random Forest, SVM, XGBoost, Voting & Stacking Ensembles, k-Fold Cross-Validation, Hypothesis Testing, Feature Selection, Hyperparameter Tuning (GridSearchCV), ROC-AUC Evaluation, Model Interpretability (SHAP)

Generative AI & Computer Vision: LLM & RAG Applications, LangChain, Embeddings, Vector Databases, YOLOv8, OpenCV

Libraries & Visualization: scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau

Data Engineering: ETL, Data Pipelines, Data Cleaning & Validation, SQL Schema Design, Analytics Workflows

Deployment & Tools: Model Deployment, Flask, Streamlit, REST APIs, Google Cloud (Vertex AI), SQLite, Firebase, Git, Jupyter Notebook

EDUCATION

Vellore Institute of Technology (VIT), Vellore *2022 – 2026*

B.Tech, Computer Science and Engineering — **CGPA: 8.36 / 10** Convocation: August 2026

CERTIFICATIONS

Oracle Cloud AI Foundations • Generative AI Applications using Vertex AI — Google Cloud

PUBLICATION

Design and Implementation of SOCIO: A Smart Community App for Participation, Safety, and Sustainable Urban Living — 10th International Conference on Internet of Things: Connected Technologies (ICIoTCT), 2025. Co-designed IoT modules for community safety monitoring and sustainable urban living.

LEADERSHIP

Senior Core Member, RoboVITics Club, VIT — Led CSE project initiatives and mentored an AI/ML team of **100+ students** through hands-on model building, evaluation, and deployment.

ACHIEVEMENTS

- **Winner, Ideathon 2025** (Innovation Council, VIT) — fleet-optimisation platform applying ride-hailing matching logic to cut truck idle time.
- **Winner, Agriculture Hackathon** — vertical-farming solution optimising water efficiency and micronutrient delivery.